Udhay Chowdhury

Computational & Experimental Bioelectronics Researcher | Organic Bioelectronics | Wearable Biosensors

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Profile

My research has evolved from perovskite solar cells to self-powered triboelectric nanogenerators and now to organic bioelectronic biosensors. I have developed a strong foundation in device physics, multiphysics simulation, and data-driven research while expanding into experimental biosensor development. I aim to integrate computational modeling with laboratory research to accelerate bioelectronic device design and optimization. My long-term goal is to develop wearable bioelectronic sensors for accessible healthcare.

Academic Credentials

B.Sc. in Electrical and Electronic Engineering — **Graduated: June 2026**

Chittagong University of Engineering & Technology (CUET), Chattogram, Bangladesh — CGPA: 3.69 / 4.00

Thesis: *MXene-Integrated Nanostructured PTFE Triboelectric Nanogenerators for Enhanced Energy Harvesting*

Relevant Coursework: Electronic Circuits, Microprocessor and Interfacing, Measurement and Instrumentation, Biomedical Instrumentation, Optoelectronics, Semiconductor Devices, Electromagnetic Fields and Waves, VLSI Design.

Research Interests

- Organic Bioelectronics
- Electrochemical Biosensors
- Wearable Biosensors
- Computational Device Physics
- Organic Electrochemical Transistors (OECT)
- Electrolyte-Gated Organic Field-Effect Transistors (EGOFET)
- Bioelectronic Devices
- Multiphysics Simulation

Research Experience

Research Assistant, STARLAB – IET, CUET

Supervisor: Adnan Faisal Siddique, Faculty Member, IET, CUET

- Developed machine-learning models (Random Forest, XGBoost and LightGBM) to predict optimal fabrication parameters for lead-free perovskite solar cells using curated experimental and simulated datasets, accelerating device optimization and improving power-conversion-efficiency prediction.

Undergraduate Thesis Research — **CUET**

- Designed and simulated MXene-integrated, nanostructured PTFE-based Triboelectric Nanogenerators (TENG) for wearable self-powered energy harvesting, achieving $\sim$300 mW/m^2 power density – a 15%+ improvement over baseline through optimized device geometry and charge-trapping layers.

Research & Development Projects

Organic Electrochemical Transistor (OECT) – 2D Device Simulation — **View Report**

- Developed a two-dimensional finite-element model of an Organic Electrochemical Transistor (OECT) in COMSOL Multiphysics by coupling Electrostatics and Transport of Diluted Species physics. Simulated Nernst–Planck ion transport, electric double-layer formation, and volumetric channel doping, reproducing transfer and output characteristics reported in the literature for low-voltage biosensor transducers.

Electrolyte-Gated Organic Field-Effect Transistor (EGOFET) Simulation — **View Report**

- Developed a multiphysics model of an Electrolyte-Gated Organic Field-Effect Transistor (EGOFET), simulating electrolyte-semiconductor coupling and electric double-layer capacitance to optimize channel geometry and transconductance for low-voltage, label-free biosensing applications.

90 nm CMOS D Flip-Flop – Schematic, Layout & Verification — **View Report**

- Designed and verified a 90 nm CMOS D flip-flop in Cadence Virtuoso (SPICE simulation, DRC/LVS checks), delivering a fully verified schematic and layout ready for system integration.

Professional Experience

Systems Analyst Intern — **June 2026 – Sept 2026**

Deployed at Grameenphone Ltd. (Telenor Group), Dhaka, Bangladesh

- Collaborated across cross-functional engineering, product, and technical teams to translate complex requirements into structured documentation (SOPs, SRS) – sharpening systematic analysis and technical-writing skills directly transferable to research documentation and protocol writing.

Industrial Training Trainee, Bangladesh Telecommunications Company Limited (BTCL)

2-week attachment, Chattogram, Bangladesh

- Observed digital switching systems and GPON-based FTTH network operation, including fault detection and configuration of OLTs/ONUs – early exposure to instrumentation and systematic fault diagnosis relevant to sensor-system troubleshooting.

Peer Reviewer, *Solar Energy* Journal **Apr 2026 – May 2026**

Freelance, Remote

- Reviewed a submitted manuscript for *Solar Energy* (Elsevier), assessing technical methodology, results, and scientific contribution; awarded a Certificate of Reviewing.

Scholarly Work

Journal Articles

- *MXene-Integrated Nanostructured PTFE Triboelectric Nanogenerators for Enhanced Energy Harvesting.* **Udhay Chowdhury** et al. Submitted (under review).
- Simulation and optimization of a $CsSnI_3/CsSnGeI_3/Cs_3Bi_2I_9$ based triple absorber layer perovskite solar cell using SCAPS-1D.
  Umme Mabrura Umama, Mohammad Iftekher Ebne Jalal, Md Adnan Faisal Siddique, **Udhay Chowdhury**, Md Inzamam Ul Hoque, Md Jahidur Rahman. *Journal of Physics and Chemistry of Solids*, Vol. 198, March 2025, Article 112480. DOI: 10.1016/j.jpcs.2024.112480

Conference Presentations

- Numerical modeling of Perovskite Tandem Solar Cell with $CH_3NH_3SnI_3$ Bottom Absorber.
  Udhay Chowdhury, Mohammad Iftekher Ebne Jalal, Md Adnan Faisal Siddique, Umme Mabrura Umama, Md Inzamam Ul Hoque, Md Jahidur Rahman. 4th ICECCE 2025, CUET, Bangladesh. DOI: 10.1109/ECCE64574.2025.11013425
- Investigation of varied Doping of ZnO as an ETL for $KSnI_3$ based Perovskite Solar Cell with ETL and Absorber Thickness Optimization.
  Md Jahidur Rahman, Md Inzamam Ul Hoque, Mohammad Iftekher Ebne Jalal, Md Adnan Faisal Siddique, **Udhay Chowdhury**, Umme Mabrura Umama. 4th ICECCE 2025, CUET, Bangladesh. DOI: 10.1109/ECCE64574.2025.11013548

Leadership

IEEE CUET Student Branch: Membership Campaign Lead (2025); Directed strategy and execution of IEEE recruitment campaign resulting in 400 new student members.
Research Camp (Biomedical Signal Processing): Coordinated EEG Signal Analysis research group.
Hult Prize CUET: Organizing Committee Member (2024–2025).

Honours & Achievements

- 1st Runners Up, "Discussion with the Changemaker" (CUET Trailblazers) – proposed a waterlogging solution for Chittagong City.
- 2nd Runners Up, "Face the Case 3.0" (Team Tetris) – business case on electric vehicles in Bangladesh.
- IEEE R10 Ethics Champion – IEEE R10 Ethics Super Power Challenge (scored $\geq$70% in ethics assessment).
- Merit scholarship for exceptional performance in Secondary School Certificate and Higher Secondary School Certificate examinations.

Technical Skills

Simulation:	COMSOL Multiphysics, SCAPS-1D, FEA, OECT, EGOFET
Programming:	Python (NumPy, Pandas, Scikit-learn, XGBoost, LightGBM), MATLAB
VLSI:	Cadence Virtuoso, SPICE, CMOS Layout, DRC/LVS
Tools:	Git/GitHub, LaTeX, Microsoft Office